

SCHOOL OF
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Soft L-Topological Language Modelling

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Team Number: AA03
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Introduction & Motivation

- Current large language models use statistical guessing using rigid Euclidean vectors on a geometric space which struggle to represent strict logic.
- We propose a shift from geometrical proximity to set inclusion using soft sets.
- To compute contextual relationships, strict mathematical logic is used, instead of dot products (self-attention).

Literature Review

- *Vaswani et al. (2017) "Attention Is All You Need"*
  - **Method:** Scaled dot-product self-attention (WMT datasets).
  - **Result/Limitation:** Established the standard, but softmax forces dense matrices (noisy weights > 0 for unrelated tokens), preventing true sparsity and logical deduction.
- *Ji et al. (2023) "Survey of Hallucination in Natural Language Generation"*
  - **Method:** Comprehensive review of standard LLM architectures.
  - **Result/Limitation:** Concludes that statistical vector approximations fail at robust polysemy resolution and factual boundary preservation.

Literature Review (Contd.)

- *Molodtsov (1999) "Soft Set Theory - First Results" / Recent Neuro-Fuzzy adaptions*
  - **Method:** Parameterized mapping of uncertainties.
  - **Result/Limitation:** Proven mathematically sound for categorization, but rarely scaled to sequence routing or dynamic NLP pipelines.
- *Pai S. S., & Baiju, T. (2021) "On Soft Lattice Topological Spaces“*
  - **Method:** Introduced Soft Lattice Topological Spaces

Research Gap

- **Absence of True Sparsity in Routing:** Standard softmax-based attention inherently passes noisy context between completely unrelated tokens. There is a lack of sequence routing mechanisms that can mathematically evaluate to exactly (true sparsity) when tokens share no context.
- **Euclidean Incompatibility with Hierarchy:** Current embeddings use Euclidean spatial distance, which cannot natively enforce asymmetrical dependencies (e.g.,). There is a gap in architectures representing language hierarchically via abstract algebra.

Problem Statement

- Current Large Language Models frequently hallucinate because they route sequential context using statistical probability and geometric proximity rather than strict logic. This project aims to design and train a novel "Soft L-Topological" neural architecture that embeds tokens as parameterized soft sets over a continuous lattice, utilizing differentiable lattice operators (t-norms and t-conorms) to compute exact logical intersections between tokens.
- **Scope:** Custom embedding layer design, replacement of self-attention with -norm/-conorm routing, scaling from initial character-level testing to a benchmarked sub-word (BPE) language model.

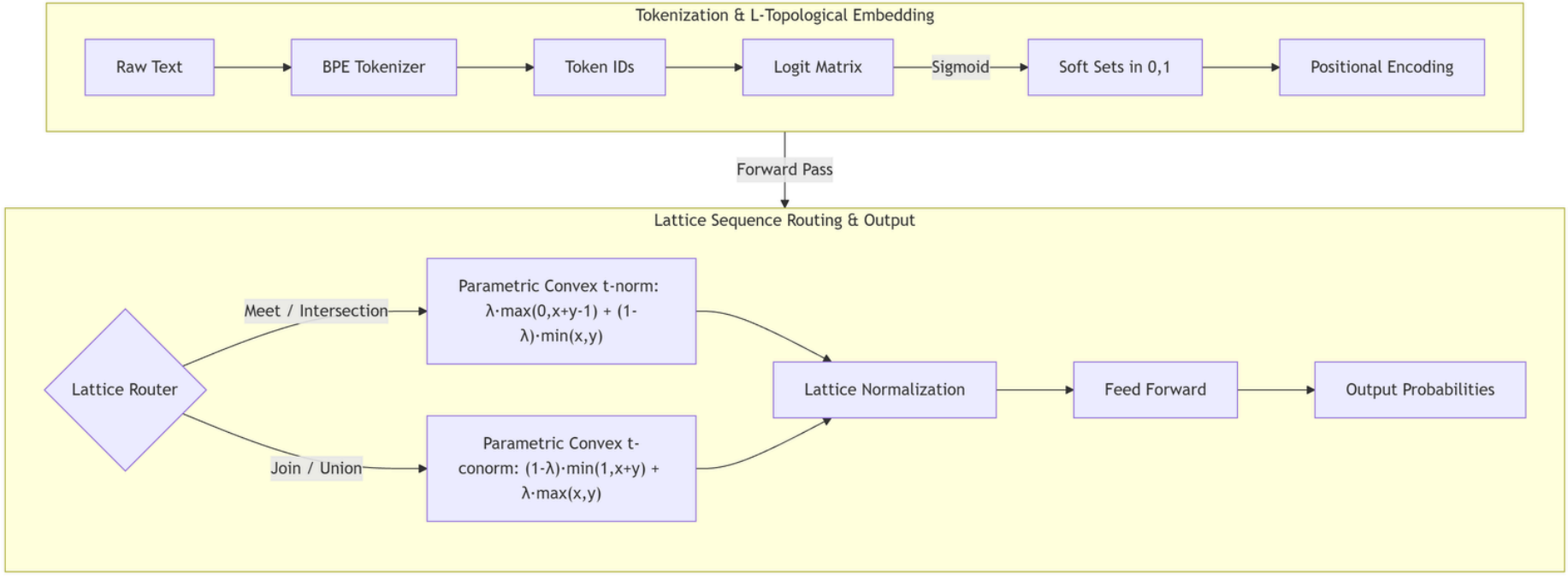
Soft L-Topology:

- **Soft: Parameterized & Context-Dependent.** Instead of evaluating items in isolation, soft sets evaluate relationships relative to specific context parameters (which act as your attention heads).
- **L: Continuous Lattice ().** Replaces binary logic with a continuous, bounded interval , making continuous formal logic fully differentiable for backpropagation.
- **Topology: Relational Containment.** Replaces spatial/geometric vector distance (dot products) with structured logical connectedness and set boundary operations.
- **Soft -Topology: Differentiable Contextual Logic.** The unified framework that combines all three—allowing attention heads (Soft) to compute exact logical intersections and unions (-norms and -conorms) over bounded features () while preserving structural relationships (Topology).

Objectives

- **Design the Embedding:** Implement a custom soft set embedding layer mapped strictly to a continuous space (using Logit-Sigmoid or Beta distribution initialization).
- **Engineer the Router:** Replace standard Multi-Head Attention with a Continuous Lattice Router utilizing mathematically differentiable -norms () and -conorms ().
- **Train & Scale the Prototype:** After unit-testing on character-level data, train a complete sub-word Language Model (10M-50M parameters) on a standard text corpus (e.g., WikiText-2).
- **Evaluate Sparsity & Hallucination:** Measure and achieve a significantly higher internal routing sparsity percentage compared to a standard dot-product baseline of identical size, demonstrating deterministic behavior.

Proposed Methodology / Framework



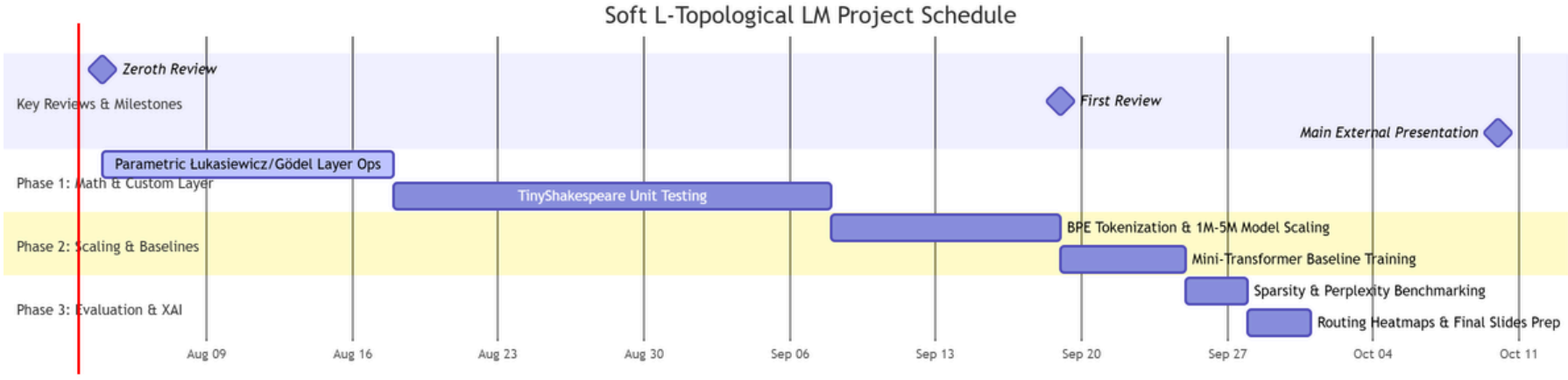
Techniques & Tools

- **Embedding Technique:** Unbounded Logit matrices squashed via Sigmoid activation to strictly enforce continuous lattice boundaries, allowing standard backpropagation.
- **Routing Algorithms:** Product t-norm () for intersection; Probabilistic Sum t-conorm () for union.
- **Datasets:** *Phase 1:* TinyShakespeare (Character-level logic unit tests). *Phase 2:* WikiText-2 / WikiText-103 (Sub-word model training and baseline benchmarking).
- **Hardware/Platform:** PyTorch / Python environment, executed on local/cloud GPUs, utilizing HuggingFace tokenizers.

Expected Outcomes

- **The Final Model:** A functional 10M-50M parameter sub-word language model capable of coherent text generation utilizing zero dot-product attention.
- **Comparative Benchmark:** Empirical data comparing our L-Topological model against a standard Transformer of exact equal size, proving superior routing sparsity (dead connections evaluating to exactly).
- **Explainable AI (XAI) Proof:** A visual output or mathematical trace showing exactly how the model arrived at a specific prediction by logging the deterministic lattice intersections (proving the mechanism that prevents hallucinations).

Timeline


Soft L-Topological LM Project Schedule
Zeroth Review
Key Reviews & Milestones
First Review
Main External Presentation
Parametric Łukasiewicz/Gödel Layer Ops
Phase 1: Math & Custom Layer
TinyShakespeare Unit Testing
BPE Tokenization & 1M-5M Model Scaling
Phase 2: Scaling & Baselines
Mini-Transformer Baseline Training
Sparsity & Perplexity Benchmarking
Phase 3: Evaluation & XAI
Routing Heatmaps & Final Slides Prep
Aug 09
Aug 16
Aug 23
Aug 30
Sep 06
Sep 13
Sep 20
Sep 27
Oct 04
Oct 11

Team Roles

Team Member	Roles
Pranav Sivakumar	• Implements PyTorch autograd ops for Łukasiewicz & Gödel t-norms. • Implements a custom soft set embedding layer.
Rithesh SS	• Integrates BPE tokenizers & scales model to 1M–5M parameters on WikiText-2. • Optimizes GPU memory usage and training throughput.
Vishwa Vinayak C K	• Builds equal-sized Transformer baseline & manages dataset splits. • Logs perplexity, loss metrics, and execution speeds.
Ashwin J R	• Computes exact routing zero-sparsity percentages. • Generates routing heatmaps for XAI proofs. • Lead editor for docs, slides, and final report.

References

- A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- Z. Ji et al., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1-38, 2023.
- D. Molodtsov, "Soft set theory—First results," *Computers & Mathematics with Applications*, vol. 37, no. 4-5, pp. 19-31, 1999.
- S. S. Pai and T. Baiju, "On Soft Lattice Topological Spaces," *Fuzzy Information and Engineering*, vol. 13, no. 1, pp. 1–16, 2021.